Chapter 12: Fluid Dynamics: Theory & Applications

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 12

Part I

Flow Rate and the Equation of Continuity

Outline of Part I: Flow Rate and the Equation of Continuity

Flow Rate

Definition: Flow Rate Q — SI Unit: m^3/s

The **flow rate** is the volume of fluid passing a point per unit time:

$$Q = \frac{V}{t}$$

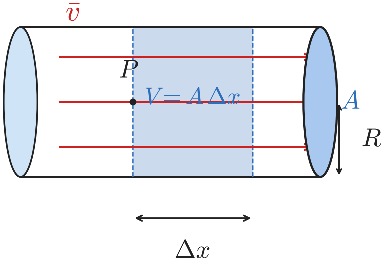
where V is volume (m^3) and t is elapsed time (s).

Note: $1\,\mathrm{L} = 10^{-3}\,\mathrm{m}^3$ and $1000\,\mathrm{L} = 1\,\mathrm{m}^3$

Flow Rate and Velocity

Imagine a pipe of cross-sectional area A and average fluid speed $\bar{v}$. In time Δt, fluid travels $\Delta x = \bar{v}\,\Delta t$, sweeping out volume $V = A\,\Delta x$:

$$Q = \frac{V}{t} = \frac{A\,\Delta x}{\Delta t} = \boxed{A\bar{v}}$$



- $Q = A\bar{v}$: flow rate = area × average speed
- For a circular pipe: $A = \pi R^2 = \pi(d/2)^2$
- Velocity is highest at the pipe center $(r = 0)$, zero at the wall $(r = R)$ — due to friction (viscosity)

Velocity Profile in a Real Pipe

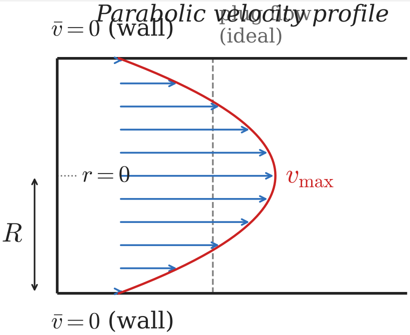
Boundary Conditions

In a real (viscous) fluid inside a pipe:

- $\bar{v}(r = R) = 0$ (fluid sticks to wall — friction)
- $\bar{v}_{\text{max}} = \bar{v}(r = 0)$ (fastest at center)
- $\left.\dfrac{d\bar{v}}{dr}\right|_{r=0} = 0$ (smooth maximum at center)

Result: a **parabolic** velocity profile.

$\bar{v}$ in the formula $Q = A\bar{v}$ is the *average* speed over the cross-section — not the centerline speed.



Types of Flow

- **Plug flow** (ideal): uniform $\bar{v}$ across entire cross-section
- **Laminar flow** (real): parabolic profile, smooth layers
- **Turbulent flow**: irregular, mixed layers — no simple profile

Example: Volume of Blood Pumped in a Lifetime

The heart pumps blood at an average flow rate of $Q = 5.00\,\mathrm{L/min}$.
How many cubic meters of blood does the heart pump in a 75-year lifetime?

$$t = 75\,\mathrm{yr} \times \frac{5.26 \times 10^5\,\mathrm{min}}{1\,\mathrm{yr}} = 3.95 \times 10^7\,\mathrm{min}$$

$$V = Qt = 5.00\,\frac{\mathrm{L}}{\mathrm{min}} \times 3.95 \times 10^7\,\mathrm{min}$$

$$V = 1.97 \times 10^8\,\mathrm{L} \times \frac{10^{-3}\,\mathrm{m}^3}{1\,\mathrm{L}}$$

$$\boxed{V = 2.0 \times 10^5\,\mathrm{m}^3}$$

$2.0 \times 10^5\,\mathrm{m}^3$ — equivalent to about 200 million liters, or roughly 80 Olympic-sized swimming pools.

The heart is a remarkably tireless pump, never stopping for the entire 75 years.

Outline of Part I: Flow Rate and the Equation of Continuity

Flow Rate

Equation of Continuity

Equation of Continuity

Conservation of Fluid: $Q_{\text{in}} = Q_{\text{out}}$

For an **incompressible** fluid (constant density — valid for liquids), the flow rate Q is the same at every cross-section along a pipe:

$$Q_1 = Q_2 \quad \Rightarrow \quad \boxed{A_1 \bar{v}_1 = A_2 \bar{v}_2}$$

As area decreases, speed increases — and vice versa.

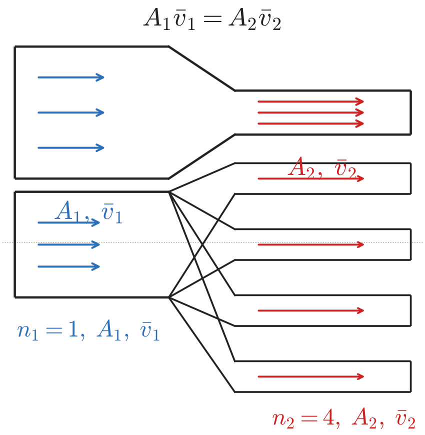
With Branching Pipes

If one large pipe branches into n_2 smaller pipes:

$$\boxed{n_1 A_1 \bar{v}_1 = n_2 A_2 \bar{v}_2}$$

where $n_1 = 1$ for a single inlet pipe.

Also: $Q_1 = Q_2 + Q_3 + Q_4 + \cdots$



Why liquids are incompressible: their atoms are already in contact. A gas compresses easily because atoms are far apart.

Example: Garden Hose and Nozzle

A garden hose (radius $r_1 = 0.900\,\text{cm}$) is fitted with a nozzle (radius $r_2 = 0.250\,\text{cm}$). Flow rate $Q = 0.500\,\text{L/s}$. Find $\bar{v}$ in (a) the hose and (b) the nozzle.

$$\text{(a)} \quad \bar{v}_1 = \frac{Q}{A_1} = \frac{Q}{\pi r_1^2}$$

$$\bar{v}_1 = \frac{5.00 \times 10^{-4}\,\frac{\text{m}^3}{\text{s}}}{\pi(9.00 \times 10^{-3}\,\text{m})^2}$$

$$\boxed{\bar{v}_1 = 1.96\,\frac{\text{m}}{\text{s}}}$$

$$\text{(b)} \quad \bar{v}_2 = \bar{v}_1 \left(\frac{r_1}{r_2}\right)^2 = 1.96\left(\frac{0.900}{0.250}\right)^2$$

$$\boxed{\bar{v}_2 = 25.5\,\frac{\text{m}}{\text{s}}}$$

The nozzle speed (25.5 m/s) is about 13× the hose speed (1.96 m/s).

Area ratio: $(r_1/r_2)^2 = (0.9/0.25)^2 = 12.96 \approx 13.$ ✓

This is exactly why a thumb over a hose increases range — higher exit speed.

Example: Blood Flow in the Cardiovascular System

(a) Find the average speed of blood in the aorta ($r_{\text{aorta}} = 10\,\text{mm}$, $Q = 5.0\,\text{L/min}$).
(b) Blood speed in capillaries is $0.33\,\frac{\text{mm}}{\text{s}}$ and capillary diameter is $8.0\,\mu\text{m}$. How many capillaries does the circulatory system have?

$$\text{(a)}\quad \bar{v}_{ao} = \frac{Q}{\pi r_{ao}^2} = \frac{8.33\times 10^{-5}\,\frac{\text{m}^3}{\text{s}}}{\pi(0.010\,\text{m})^2}$$

$$\boxed{\bar{v}_{ao} = 0.27\,\frac{\text{m}}{\text{s}}}$$

$$\text{(b)}\quad n_2 = \frac{n_1 A_{ao}\bar{v}_{ao}}{A_{cap}\bar{v}_{cap}}$$

$$n_2 = \frac{(1)\pi(0.010\,\text{m})^2(0.27\,\frac{\text{m}}{\text{s}})}{\pi(4.0\times 10^{-6}\,\text{m})^2(3.3\times 10^{-4}\,\frac{\text{m}}{\text{s}})}$$

$$\boxed{n_2 \approx 5.0\times 10^9\ \text{capillaries}}$$

5 billion capillaries! The enormous total cross-sectional area slows blood to $\sim$0.33 mm/s — slow enough for gas and nutrient exchange without clotting.

Continuity explains why: same Q, vastly larger total $A \Rightarrow$ much smaller $\bar{v}$.

Check Your Understanding

Water flows through a pipe of diameter 4.0 cm at a speed of $2.0\,\frac{\mathrm{m}}{\mathrm{s}}$. The pipe narrows to a diameter of 2.0 cm.

(a) What is the flow rate Q?

(b) What is the water speed in the narrow section?

Come to lecture to see the answer.

Part II

Bernoulli's Equation

Outline of Part II: Bernoulli's Equation

Bernoulli's Equation: Energy Conservation for Fluids

Bernoulli's Equation

For an incompressible, frictionless (ideal) fluid, energy is conserved along a streamline:

$$\boxed{P_1 + \frac{1}{2}\rho\bar{v}_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho\bar{v}_2^2 + \rho g h_2}$$

Each Term: Energy per Unit Volume

P	pressure (work/volume)
$\frac{1}{2}\rho\bar{v}^2$	kinetic energy/volume
$\rho g h$	gravitational PE/volume

Units: $\dfrac{\mathrm{J}}{\mathrm{m}^3} = \dfrac{\mathrm{N}}{\mathrm{m}^2} = \mathrm{Pa}$ ✓

Special Cases

- **Fluid at rest** ($\bar{v} = 0$): recovers $P_2 = P_1 + \rho g h$ (hydrostatics)
- **Constant height** ($h_1 = h_2$): $P + \frac{1}{2}\rho\bar{v}^2 = \text{const}$
- **Bernoulli's Principle**: at constant height, *pressure drops where speed increases*

Bernoulli's Principle: Faster Flow ⇒ Lower Pressure

At *constant height*: as fluid speed increases, pressure decreases.

$$P_1 + \tfrac{1}{2}\rho\bar{v}_1^2 = P_2 + \tfrac{1}{2}\rho\bar{v}_2^2 \quad \Rightarrow \quad P_1 = P_2 + \tfrac{1}{2}\rho(\bar{v}_2^2 - \bar{v}_1^2)$$

Applications of Bernoulli's Principle

- **Airplane lift**: air flows faster over the curved upper wing surface ⇒ lower pressure above wing ⇒ net upward (lift) force
- **Sailboat sail**: fast air on leeward side ⇒ lower pressure ⇒ forward thrust
- **Atomizer/perfume spray**: fast air jet above a tube ⇒ low pressure draws up liquid
- **Trucks passing cars**: fast air between vehicles ⇒ low pressure draws vehicles together (a real driving hazard)
- **Roofs blowing off**: fast wind above a roof ⇒ lower pressure above ⇒ net upward force

Example: Pressure Drop in a Garden Hose

Using the hose/nozzle from before: $\bar{v}_1 = 1.96\,\frac{\mathrm{m}}{\mathrm{s}}$ (hose), $\bar{v}_2 = 25.5\,\frac{\mathrm{m}}{\mathrm{s}}$ (nozzle exit at $P_2 = P_{\text{atm}} = 1.01 \times 10^5\,\mathrm{Pa}$). Find the pressure in the hose P_1. Assume level flow.

$$P_1 = P_2 + \tfrac{1}{2}\rho(\bar{v}_2^2 - \bar{v}_1^2)$$

$$P_1 = 1.01 \times 10^5 + \tfrac{1}{2}(1000)\left[(25.5)^2 - (1.96)^2\right]$$

$$P_1 = 1.01 \times 10^5 + \tfrac{1}{2}(1000)(650.25 - 3.84)$$

$$P_1 = 1.01 \times 10^5 + 3.23 \times 10^5$$

$$\boxed{P_1 = 4.24 \times 10^5\,\mathrm{Pa} \approx 4.19\,\mathrm{atm}}$$

The hose must be at 4.2 atm — much higher than atmospheric.

This high pressure is what *accelerates* the water to 25.5 m/s in the nozzle. The nozzle exits at atmospheric pressure (into open air).

Torricelli's Theorem

Exit Speed from a Large Reservoir

A large open tank has a hole at depth h below the surface. Both surface and hole are open to the atmosphere ($P_1 = P_2 = P_{\text{atm}}$). The surface drops slowly ($\bar{v}_1 \approx 0$):

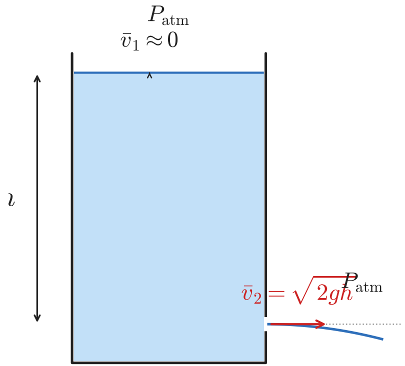
$$\frac{1}{2}\rho\bar{v}_1^2 + \rho g h_1 = \frac{1}{2}\rho\bar{v}_2^2 + \rho g h_2$$

$$\rho g h = \frac{1}{2}\rho\bar{v}_2^2 \quad \Rightarrow \quad \boxed{\bar{v}_2 = \sqrt{2gh}}$$

Torricelli's Theorem: the exit speed equals the speed a free-falling object would reach after falling from the same height h.

Example: Water exits a hole 5.0 m below the surface:

$$\bar{v} = \sqrt{2(9.80)(5.0)} = 9.9\ \frac{\text{m}}{\text{s}}$$



Real-world examples:

- Water gushing from the base of a dam
- Leak at the bottom of a water tank
- Speed of blood from a cut artery

Power in Fluid Flow

Power Delivered by a Moving Fluid

Multiply Bernoulli's energy per unit volume by Q (volume per unit time):

$$\text{Power} = \left(P + \tfrac{1}{2}\rho\bar{v}^2 + \rho g h\right) Q$$

Three contributions to power at any cross-section:

$$\text{Power}_{\text{work}} = P\,Q \qquad \text{(pressure does work on fluid)}$$

$$\text{Power}_{KE} = \tfrac{1}{2}\rho\bar{v}^2\,Q \qquad \text{(kinetic energy flux)}$$

$$\text{Power}_{PE} = \rho g h\,Q \qquad \text{(potential energy flux)}$$

A pump increases the total energy per unit fluid volume. The pump power needed is:

$$P_{\text{pump}} = \Delta P \cdot Q = (P_{\text{out}} - P_{\text{in}}) \cdot Q$$

if the pump only changes pressure (no change in speed or height).

Example: Fire Hose Pump Power

A fire truck pump boosts water pressure from a hydrant supply of $P_{\text{in}} = 0.700 \times 10^6\,\text{Pa}$ to $P_{\text{out}} = 1.62 \times 10^6\,\text{Pa}$ at a flow rate of $Q = 40.0\,\text{L/s}$. Find the pump's output power.

$$\Delta P = P_{\text{out}} - P_{\text{in}}$$

$$\Delta P = 1.62 \times 10^6 - 0.700 \times 10^6 = 0.920 \times 10^6\,\text{Pa}$$

$$\text{Power} = \Delta P \cdot Q$$

$$\text{Power} = (9.20 \times 10^5\,\text{Pa})(4.00 \times 10^{-2}\,\tfrac{\text{m}^3}{\text{s}})$$

$$\boxed{\text{Power} = 3.68 \times 10^4\,\text{W} = 36.8\,\text{kW}}$$

$36.8\,\text{kW} \approx 49$ horsepower — consistent with the large pumps on fire trucks.

The heart is an analogous pump. At rest, the heart outputs $\sim 1.3\,\text{W}$ — small, but continuous for 75 years.

Check Your Understanding

Roofs are sometimes pushed *off vertically* during a strong hurricane or tornado, even when the windows are shut.

Use Bernoulli's principle to explain this phenomenon.

Come to lecture to see the answer.

Check Your Understanding

Water flows from a large open tank through a hole at depth $h = 2.00\,\mathrm{m}$ below the surface.

(a) What is the exit speed of the water?

(b) If the hole has area $A = 1.00\,\mathrm{cm}^2$, what is the flow rate?

Come to lecture to see the answer.

Part III

Viscosity and Laminar Flow

Outline of Part III: Viscosity and Laminar Flow

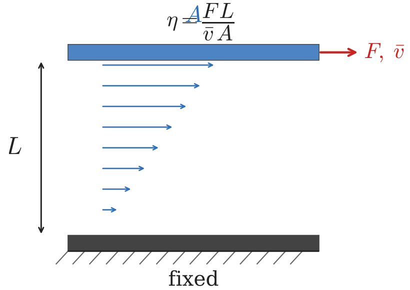
Viscosity: A Fluid's Internal Friction

Definition: Viscosity η — SI Unit: Pa·s

Viscosity quantifies a fluid's resistance to flow due to internal friction between fluid layers. Think of it as the "thickness" or "stickiness" of a fluid.

$$\boxed{\eta = \frac{F\,L}{\bar{v}\,A}}$$

where F = shear force (N), L = layer separation (m), $\bar{v}$ = relative speed of layers (m/s), A = area (m^2).



Selected Viscosity Values

Fluid	Temp	η (Pa·s)
Water	20°C	1.00×10^{-3}
Blood	37°C	2.08×10^{-3}
Blood plasma	37°C	1.81×10^{-3}
Motor oil	20°C	~ 0.2
Air	20°C	1.81×10^{-5}

- Liquids: viscosity *decreases* with temperature (weaker cohesion)
- Gases: viscosity *increases* with temperature (more collisions between molecules)

Poiseuille's Law: Flow in a Tube

Poiseuille's Law for Laminar Flow in a Cylinder

For a viscous fluid flowing through a cylindrical tube of length l and radius r:

$$\boxed{Q = \frac{(P_2 - P_1)\,\pi r^4}{8\eta\, l}}$$

Equivalently, defining **resistance** $R = \dfrac{8\eta\, l}{\pi r^4}$:

$$Q = \frac{P_2 - P_1}{R} \qquad \text{(analogous to Ohm's Law: } I = V/R\text{)}$$

Flow rate depends on r^4: **doubling the radius increases flow 16-fold**

Resistance $R \propto 1/r^4$: a small narrowing of an artery dramatically increases resistance.

Understanding Poiseuille's Law

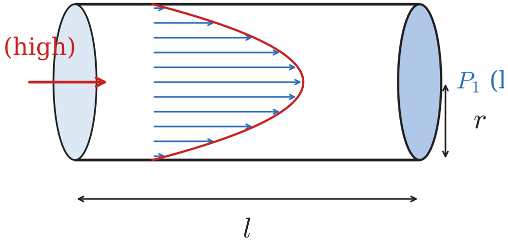
What Increases Flow Rate Q?

- Larger $\Delta P = P_2 - P_1$ (more driving pressure)
- Larger radius r (drastically — $Q \propto r^4$)
- Shorter tube length l
- Lower viscosity η

Analogy to electrical circuits:

- $Q \leftrightarrow I$ (current)
- $\Delta P \leftrightarrow V$ (voltage)
- $R \leftrightarrow R$ (resistance)
- Fluids flow from high P to low P; current flows from high V to low V

$$Q = \frac{(P_2 - P_1)\,\pi r^4}{8\eta l}$$



General Q–ΔP Relationship

For pipe network with no elevation change:

$$Q \propto \Delta P \quad Q \propto \frac{1}{R}$$

Large R (long or narrow pipe) $\Rightarrow$ small Q for same ΔP.
Small R (short or wide pipe) $\Rightarrow$ large Q.

Example: Plaque and Artery Narrowing

Plaque deposits narrow an artery so that blood flow is reduced to *half* its normal value. Assuming laminar flow, constant ΔP, and constant l, by what factor has the radius been reduced?

$$Q \propto r^4 \quad \Rightarrow \quad \frac{Q_2}{Q_1} = \left(\frac{r_2}{r_1}\right)^4$$

$$0.5 = \left(\frac{r_2}{r_1}\right)^4$$

$$\frac{r_2}{r_1} = (0.5)^{1/4} = \boxed{0.841}$$

The radius was reduced by only **16**%, yet blood flow was cut in half.

This is why arterial plaque is so dangerous: a seemingly small narrowing causes severe flow reduction.

Conversely, medications that dilate arteries by just 16% can double blood flow — explaining how vasodilators treat hypertension.

Example: IV Needle Pressure

Find the gauge pressure needed to force saline through an IV needle: $Q = 1.20 \times 10^{-7}\,\frac{\mathrm{m}^3}{\mathrm{s}}$, needle radius $r = 0.150\,\mathrm{mm}$, needle length $l = 2.50\,\mathrm{cm}$, vein gauge pressure $P_1 = 1066\,\mathrm{Pa}$ (8.00 mm Hg), $\eta_{\mathrm{saline}} = 1.00 \times 10^{-3}\,\mathrm{Pa\,s}$.

$$P_2 = \frac{8\eta l Q}{\pi r^4} + P_1$$

$$P_2 = \frac{8(1 \times 10^{-3})(2.5 \times 10^{-2})(1.2 \times 10^{-7})}{\pi(1.50 \times 10^{-4})^4} + 1066$$

$$\boxed{P_2 = 1.62 \times 10^4\,\mathrm{Pa} \approx 122\,\mathrm{mmHg}}$$

Achieved by hanging the IV bag $\sim$1.6 m above the needle:

$$h = \frac{P_2}{\rho g} \approx 1.6\,\mathrm{m}$$

This is why IV poles have a specific height.

Check Your Understanding

Two pipes carry water with the same pressure difference and the same viscosity.

Pipe A has radius r and length l.
Pipe B has radius $2r$ and length $2l$.

What is the ratio Q_B/Q_A?

Come to lecture to see the answer.

Check Your Understanding

When you flush a toilet while someone is showering, the water in the shower often suddenly becomes very hot or very cold.

Use Poiseuille's Law and what you know about plumbing to explain why.

Come to lecture to see the answer.

Part IV

Turbulence and Viscous Motion

Outline of Part IV: Turbulence and Viscous Motion

The Onset of Turbulence

Motion of an Object in a Viscous Fluid

Laminar vs. Turbulent Flow

Laminar Flow

- Fluid moves in smooth, parallel layers (laminae)
- Layers do *not* mix
- Velocity profile is parabolic in a tube
- Poiseuille's Law applies
- Streamlines are smooth and continuous

Turbulent Flow

- Fluid develops eddies, swirls, and irregular motion
- Layers mix chaotically
- Poiseuille's Law does *not* apply
- Much greater resistance — energy is wasted in eddies
- Inherently chaotic: sensitive to initial conditions

Turbulent flow requires *more* pressure difference to maintain the same flow rate as laminar flow, because energy is lost to eddies and mixing. This is why turbulent blood flow (e.g., in a narrowed artery) places extra strain on the heart.

Reynolds Number: Predicting Flow Type

Reynolds Number N_R for Flow in a Tube

A dimensionless number that predicts whether flow is laminar or turbulent:

$$\boxed{N_R = \frac{2\rho\bar{v}r}{\eta}}$$

where ρ = fluid density (kg/m^3), $\bar{v}$ = average speed (m/s), r = tube radius (m), η = viscosity (Pa·s).

Critical Values

N_R	Flow type
< 2000	Laminar
$2000 - 3000$	Transitional
> 3000	Turbulent

N_R represents the ratio of *inertial forces* (which destabilize flow and cause turbulence) to *viscous forces* (which dampen disturbances and maintain laminar flow).

High viscosity $\Rightarrow$ low N_R $\Rightarrow$ laminar.
High speed or low viscosity $\Rightarrow$ high N_R $\Rightarrow$ turbulent.

Example: Is the IV Flow Laminar?

Verify that the flow through the IV needle from the previous example is indeed laminar.
Given: $Q = 1.20 \times 10^{-7}\,\frac{\mathrm{m}^3}{\mathrm{s}}$, $r = 1.50 \times 10^{-4}\,\mathrm{m}$, $\rho_{\mathrm{saline}} = 1025\,\frac{\mathrm{kg}}{\mathrm{m}^3}$, $\eta = 1.00 \times 10^{-3}\,\mathrm{Pa\,s}$.

$$A = \pi r^2 = \pi(1.50 \times 10^{-4})^2 = 7.07 \times 10^{-8}\,\mathrm{m}^2$$

$$\bar{v} = \frac{Q}{A} = \frac{1.20 \times 10^{-7}}{7.07 \times 10^{-8}} = 1.70\,\frac{\mathrm{m}}{\mathrm{s}}$$

$$N_R = \frac{2\rho\bar{v}r}{\eta} = \frac{2(1025)(1.70)(1.50 \times 10^{-4})}{1.00 \times 10^{-3}}$$

$$\boxed{N_R = 523}$$

$N_R = 523 \ll 2000 \Rightarrow$ **laminar flow**. ✓

The Poiseuille's Law calculation in the previous example was valid.

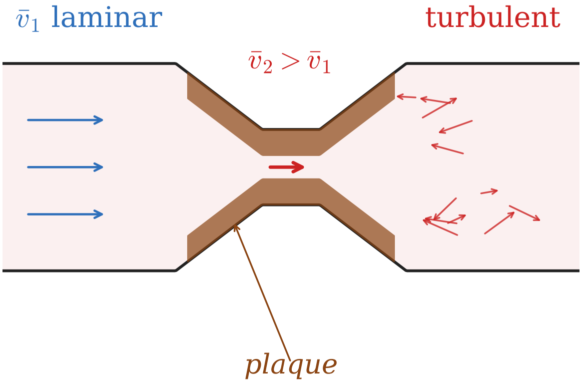
IV fluids always flow in the laminar regime due to the small needle radius and slow flow.

Turbulence in the Cardiovascular System

Normal resting aortic blood flow is **laminar**, not turbulent.

Pathological Turbulence

- **Arterial stenosis (plaque)**: narrowing increases $\bar{v}$, driving N_R above 3000 in the constriction, creating turbulent flow downstream
- **Heart valve disease**: defective valves allow regurgitation at high speed
- Both produce audible sounds (murmurs/bruits) detectable by stethoscope



Korotkoff sounds:
Turbulent-blood-flow sounds heard through a stethoscope during a blood-pressure measurement.

Appear when cuff pressure = systolic; disappear at diastolic.

Check Your Understanding

Blood ($\rho = 1050\,\frac{\text{kg}}{\text{m}^3}$, $\eta = 2.08 \times 10^{-3}\,\text{Pa s}$) flows through the aorta ($r = 0.010\,\text{m}$) at $\bar{v} = 0.27\,\frac{\text{m}}{\text{s}}$ at rest, and up to $\bar{v} = 1.0\,\frac{\text{m}}{\text{s}}$ during vigorous exercise.

Calculate N_R at rest and during exercise. Is aortic flow laminar or turbulent in each case?

Come to lecture to see the answer.

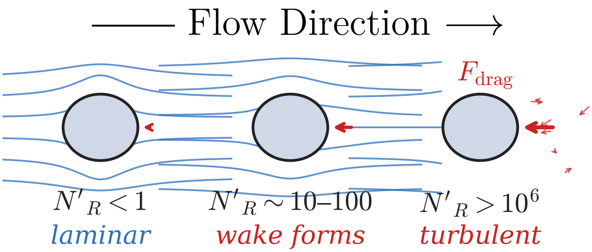
Outline of Part IV: Turbulence and Viscous Motion

Reynolds Number for Objects Moving Through Fluid

Modified Reynolds Number N'_R

For an *object* moving through a fluid (rather than fluid in a tube), the Reynolds number uses the object's characteristic length L (e.g., sphere diameter):

$$N'_R = \frac{\rho\,\bar{v}\,L}{\eta}$$



Flow Regimes Around an Object

N'_R	Flow characteristics
< 1	Laminar (smooth streamlines)
1–10	Wake begins to form
10–10^6	Partial turbulent wake
$> 10^6$	Fully turbulent wake

Stokes' Law (Laminar Regime)

For a small sphere $(N'_R < 1)$:

$$F_S = 6\pi R\eta\bar{v}$$

R = sphere radius, η = viscosity, $\bar{v}$ = speed

Terminal Speed in a Viscous Fluid

Force Balance at Terminal Speed

A sphere falling through a viscous fluid reaches terminal speed $\bar{v}_t$ when $\vec{F}_{\text{net},y} = 0$:

$$\Rightarrow \quad w = F_S + F_B \;\Rightarrow\; mg = 6\pi R\eta\bar{v}_t + \rho_{\text{fl}} Vg$$

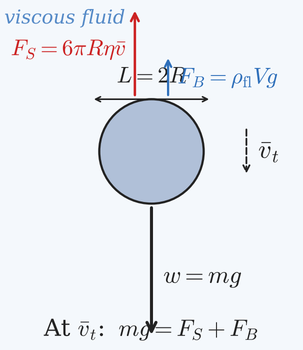
Example: Reynolds Number for a Baseball

Baseball: diameter $d = 7.40\,\text{cm}$, thrown at $\bar{v} = 40.0\,\frac{\text{m}}{\text{s}}$ through air ($\rho = 1.29\,\frac{\text{kg}}{\text{m}^3}$, $\eta = 1.81\times10^{-5}\,\text{Pa s}$):

$$N_R' = \frac{\rho\bar{v}L}{\eta} = \frac{(1.29)(40.0)(0.0740)}{1.81\times10^{-5}}$$

$$\boxed{N_R' = 2.11\times10^5}$$

A large turbulent wake forms behind the baseball, which pitchers exploit with spin to create curveballs and sliders.



Applications of viscous-fluid terminal speed:

- Sedimentation of blood cells
- Measurement of particle size in fluids
- Raindrop fall speed
- Skydiver before parachute deployment

Check Your Understanding

A helium balloon is tied inside a car. When the car accelerates forward, does the balloon move toward the front or the back of the car?

(Recall that air inside the car is denser than helium, and think about what happens to the air when the car accelerates.)

Come to lecture to see the answer.

Part V

Molecular Transport: Diffusion and Osmosis

Outline of Part V: Molecular Transport: Diffusion and Osmosis

Diffusion, Osmosis, and Related Processes

Diffusion: Molecular Transport Without Bulk Flow

Diffusion is the net transport of molecules from a region of *high concentration* to a region of *low concentration*, driven by random thermal motion — no bulk fluid flow required.

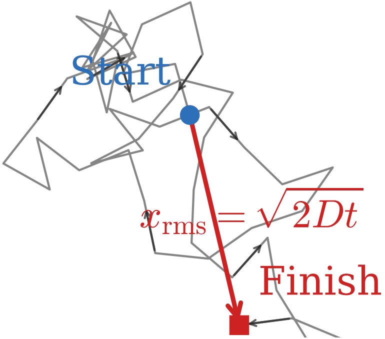
RMS Diffusion Distance

The root-mean-square distance traveled by a diffusing molecule in time t:

$$\boxed{x_{\text{rms}} = \sqrt{2Dt}}$$

where D is the **diffusion constant** (m^2/s), a property of the molecule and medium.

Note: $x_{\text{rms}} \propto \sqrt{t}$ — doubling the distance requires *quadrupling* the time.



A molecule undergoes a **random walk**, $\because$ constant collisions redirect it randomly.

Net displacement grows as $\sqrt{t}$, not linearly with t.

Diffusion Constants

Molecule	**Medium**	D (m^2/s)	**Notes**
H_2	Air	6.4×10^{-5}	Very light, fast
O_2	Air	1.8×10^{-5}	Oxygen in air
O_2	Water	1.0×10^{-9}	10,000× slower than in air
Glucose	Water	6.7×10^{-10}	Small sugar molecule
Hemoglobin	Water	6.9×10^{-11}	Large protein
DNA	Water	1.3×10^{-12}	Giant molecule

D increases with temperature (faster thermal motion) and decreases with increasing molecular mass (heavier molecules move more slowly at the same temperature). Diffusion in water is $\sim$10,000× slower than in air.

Example: How Long Does Glucose Take to Diffuse 1 cm?

A glucose molecule in water ($D = 6.7 \times 10^{-10}\,\frac{\mathrm{m}^2}{\mathrm{s}}$) must diffuse $x = 1.0\,\mathrm{cm}$. How long does this take?

$$x_{\mathrm{rms}} = \sqrt{2Dt} \quad \Rightarrow \quad t = \frac{x_{\mathrm{rms}}^2}{2D}$$

$$t = \frac{(0.010\,\mathrm{m})^2}{2(6.7 \times 10^{-10}\,\frac{\mathrm{m}^2}{\mathrm{s}})}$$

$$t = \frac{1.0 \times 10^{-4}}{1.34 \times 10^{-9}}$$

$$\boxed{t = 7.5 \times 10^4\,\mathrm{s} \approx 21\,\mathrm{h}}$$

$t = 21$ hours to diffuse 1 cm

This is why the circulatory system exists: the body cannot rely on diffusion for macroscopic transport. Blood flow delivers nutrients rapidly; capillaries bring blood within $\sim 40\,\mu\mathrm{m}$ of every cell, where diffusion then covers the final short distance ($\sim$seconds).

Osmosis and Dialysis

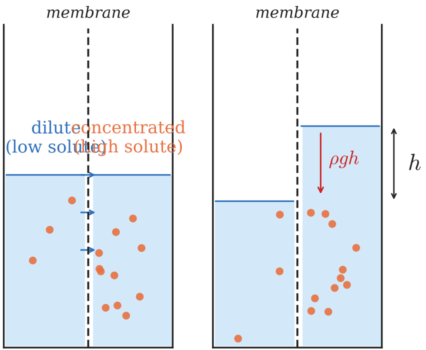
Osmosis

Transport of **solvent** (water) through a *semipermeable membrane* from high water concentration (low solute) to low water concentration (high solute).

Osmosis continues until the back pressure $\rho g h$ (called **osmotic pressure**) equals the concentration-driven force.

Dialysis

Transport of **solute** (not solvent) through a semipermeable membrane from high to low concentration. Used in kidney dialysis machines to remove waste from blood.



(a) initial *(b) equilibrium*

Osmosis vs. Dialysis:

- Both: semipermeable membranes + concentration gradients
- Both driven by concentration differences (no energy input)
- **Osmosis**: water (solvent) moves
- **Dialysis**: solute molecules move

Active Transport and Biological Membranes

Active Transport

Unlike diffusion and osmosis (which require no energy input), **active transport** uses metabolic energy (ATP) to move substances *against* a concentration gradient.

Biological Examples

- **Na^+/K^+ pump**: maintains ionic concentration differences across nerve and muscle cell membranes — essential for nerve impulses and muscle contraction
- **Glucose uptake**: cells actively import glucose even when intracellular glucose is already high
- **Intestinal absorption**: nutrients are pumped from the intestine into the bloodstream against concentration gradients

Active transport consumes approximately **25% of the body's total metabolic energy** — a surprisingly large fraction, illustrating how much energy is required simply to maintain the body's chemical gradients.

Check Your Understanding

(a) Why does the rate of diffusion increase with temperature?

(b) How are osmosis and dialysis similar? How do they differ?

(c) Could diffusion alone supply oxygen to cells 1 mm from a capillary ($D_{O_2 \text{ in water}} = 1.0 \times 10^{-9}\ \frac{\text{m}^2}{\text{s}}$)? Estimate the time.

Come to lecture to see the answer.

Chapter Summary

Key Equations — Chapter 12

Concept	Equation	Note
Flow rate	$Q = V/t = A\bar{v}$	m^3/s
Continuity	$A_1\bar{v}_1 = A_2\bar{v}_2$	incompressible
Branching	$n_1 A_1\bar{v}_1 = n_2 A_2\bar{v}_2$	
Bernoulli	$P_1 + \frac{1}{2}\rho\bar{v}_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho\bar{v}_2^2 + \rho g h_2$	ideal fluid
Torricelli	$\bar{v} = \sqrt{2gh}$	large reservoir
Power in flow	$\text{Power} = (P + \frac{1}{2}\rho\bar{v}^2 + \rho g h)\, Q$	
Viscosity	$\eta = FL/(vA)$	Pa·s
Poiseuille's Law	$Q = (P_2 - P_1)\pi r^4/(8\eta l)$	laminar
Resistance	$R = 8\eta l/(\pi r^4)$	$Q = \Delta P/R$
Reynolds (tube)	$N_R = 2\rho\bar{v}r/\eta$	< 2000: laminar
Reynolds (object)	$N'_R = \rho\bar{v}L/\eta$	
Stokes' drag	$F_S = 6\pi R\eta\bar{v}$	$N'_R < 1$
Diffusion	$x_{\text{rms}} = \sqrt{2Dt}$	